

Multi-period Market Competition

Silvana Krasteva

ECON630
Texas A&M University

April 4, 2012

Models of Multi-period Dynamic Competition

- Sequential price/quantity choice by competing firms;
 - Stackelberg game;
- Entry models:
 - Simultaneous entry choice by firms;
 - Incumbent-Entrant Models;
 - strategic choices to affect future competition;

Sequential quantity setting game: Stackelberg

- Two firms: one leader (L) and one follower (F).
- Inverse demand is $p(q) = a - bq$ where $q = q_L + q_F$.
- Constant MC $c \geq 0$ for both firms.
- Timing of the game:
 - 1 Firm L moves first and chooses its quantity q_L .
 - 2 Firm F observes q_L and chooses q_F .
- We can derive the SPNE by employing backward induction.

Stage 2

- The follower observes q_L and chooses q_F to maximize his profits:

$$\max_{q_F} (a - bq_L - bq_F - c)q_F$$

$$r_F(q_L) = \frac{a - bq_L - c}{2b}$$

- For SPNE, we require that the leader can only hold beliefs about F's strategy that is consistent with $r_F(q_L)$.
 - Note: Cournot output is a NE here, but not SPNE.

Stage 1

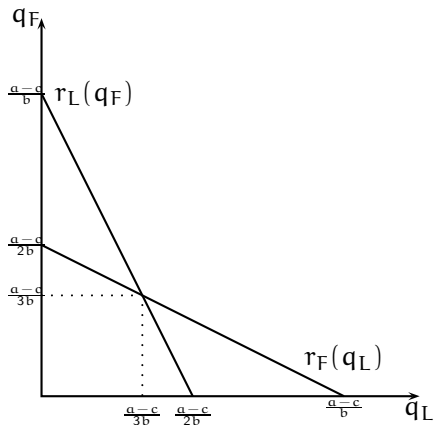
- The leader's optimization problem in the first stage is

$$\max_{q_L} (a - bq_L - br_F(q_L) - c)q_L$$

$$\max_{q_L} \frac{a - bq_L - c}{2} q_L$$

$$q_L^* = \frac{a - c}{2b}, q_F^* = \frac{a - c}{4b}$$

Stackelberg v.s Cournot



- Output:

$$q_L^S = q_L^M > q_L^C$$

$$q_F^S < q_F^C$$

$$q^S > q^C$$

- Profits:

$$\pi_L^C < \pi_L^S < \pi_L^M$$

$$\pi_F^S < \pi_F^C$$

Market Entry

- So far, number of firms has been exogenous.
- Endogenous entry: entry will occur as long as set-up costs K are less than potential profits.
- Consider an inverse demand function $p(q) = a - bq$.
- Constant MC $c \geq 0$.
- Timing:
 - 1 All potential firms simultaneously decide in or out where in is associated with a set-up cost K .
 - 2 All entrants J play a Cournot game.
- We solve the game by backward induction.

Stage 2

- Suppose that J firms have entered the market.
- Firm i 's optimization problem under Cournot is as follows:

$$\max_{q_i} (a - b \sum_{j=1}^J q_j - c) q_i$$

$$r_i(q_{-i}) = \frac{a - b \sum_{j \neq i} q_j - c}{2b}$$

- Symmetric NE:

$$q_i^* = \frac{a - c}{b(J + 1)}$$

$$\pi_i^* = \frac{1}{b} \left(\frac{a - c}{J + 1} \right)^2$$

Stage 1: Entry

- Let J^* be the SPNE entrants in the market.
- No deviation incentives requires

$$\pi^*(J) \geq K \quad (1)$$

$$\pi^*(J+1) < K \quad (2)$$

- From our Cournot example, firm J is exactly indifferent between entry and staying out if

$$(J+1)^2 = \frac{(a-c)^2}{bK}$$

- Then $J^* = \lfloor \frac{(a-c)}{\sqrt{bK}} - 1 \rfloor$

Socially optimal number of firms

- We want to maximize the total surplus w.r.t. J :

$$TS = PS + CS - KJ = J \frac{(a-c)^2}{b(J+1)^2} + J^2 \frac{(a-c)^2}{2b(J+1)^2} - JK$$

- Differentiating w.r.t. J ,

$$\frac{(a-c)^2}{b(J+1)^3} - K = 0$$

$$(J+1)^3 = \frac{(a-c)^2}{bK}$$

- Let $\tilde{J} = \lfloor \frac{(a-c)^{2/3}}{(bK)^{1/3}} - 1 \rfloor$.
- $\tilde{J} < J^*$ - there is too much entry in SPNE!

Strategic Investment

- Incumbent firms can affect future competition.
- Example: investment in cost reducing technology.
 - Firm L can reduce its MC according to $c_L(I)$ where I denotes the level of investment.
 - $c'_L(I) < 0$.
 - Cost of investment $K(I)$ with $K' > 0$ and $K'' > 0$.
 - Firm F has access to MC c_F where $c_F = c_L(0)$.
 - The market demand is $p(q) = a - q$.
 - Timing:
 - 1 L chooses I and $c_L(I)$ is realized.
 - 2 F chooses whether to enter or not.
 - 3 If entry took place, the two firms compete with costs c_L and c_F respectively. If no entry took place, L is a monopolist.

Stage 3

- If L is a monopolist, he maximizes:

$$\max_{q_L} (a - q_L - c_L)q_L$$

$$q_L^M = \frac{a - c_L}{2}, \pi_L^M = \frac{(a - c_L)^2}{4}$$

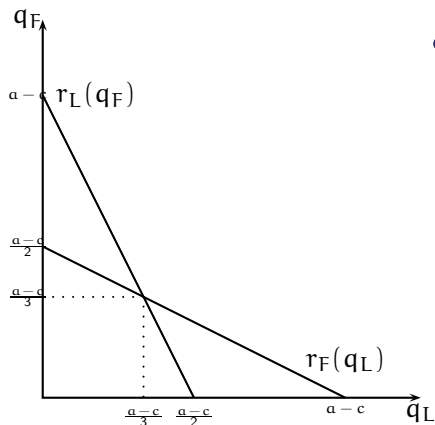
- If L and F compete, then firm i's profit function is:

$$\max q_i (a - q_i - q_j - c_i)q_i$$

$$r_i(q_j) = \frac{a - q_j - c_i}{2}$$

$$q_i^* = \frac{a + c_j - 2c_i}{3}, \pi_i^* = \frac{(a + c_j - 2c_i)^2}{9}$$

Effect of lower c_i



- There is both direct and strategic effect of a decrease in c_i on π_i^* .

$$\begin{aligned} \frac{d\pi_i^*(q_i^*, q_j^*, c_i)}{dc_i} &= \frac{d\pi_i^*}{dc_i} + \\ &+ \frac{d\pi_i^*}{dq_i^*} \frac{dq_i^*}{dc_i} + \\ &+ \frac{d\pi_i^*}{dq_j^*} \frac{dq_j^*}{dc_i} \end{aligned}$$

- The direct and the strategic effect reinforce each other.

Stage 2: Entry choice by F

- Firm F enters as long as:

$$q_F^* = \frac{a + c_L - 2c_F}{3} > 0 \implies c_L > 2c_F - a$$

- Two cases:
 - $a > 2c_F$ -entry always takes place for all c_L .
 - $a \leq 2c_F$ - F stays out for $c_L > 2c_F - a$.

Investment Stage

- Let $c_L(I) = \max\{c_F - I, 0\}$ and $K(I) = \frac{\beta I^2}{2}$ with $\beta \geq 1$.
- Consider two cases:
 - $a > 2c_F$
 - Then the profit for the leader and the follower is:

$$\pi_L^*(I) = \frac{(a - c_F + 2I)^2}{9}, \quad \pi_F^*(I) = \frac{(a - c_F - I)^2}{9}$$

- Leader's optimization:

$$\max_I \frac{(a - c_F + 2I)^2}{9} - \frac{\beta I^2}{2}$$

$$I^* = \min\left\{\frac{4(a - c_F)}{9\beta - 8}, c_F\right\}$$

Investment Stage

- Case 2:
 - $a \leq 2c_F$
 - It is sufficient to set $I = a - c_F$ to force the follower out of the market.
 - Note: $a - c_F \geq I^M = \frac{2(a - c_F)}{4\beta - 2}$.
 - Therefore, L chooses between $I_1 = \frac{4(a - c_F)}{9\beta - 8}$ and $I_2 = a - c_F$ by comparing the respective profits.

Optimal I^*

There exists $\tilde{\beta} \in (1, 2)$ such that if $\beta \in [1, \tilde{\beta}]$, the firm sets $I^* = I_2$ resulting in a monopoly in the downstream. For $\beta > \tilde{\beta}$, the firm sets $I^* = I_1$ resulting in a downstream Cournot competition with $c_L < c_F$.

Further Questions to Consider

- What is the socially optimal investment level and how it compares to I^* ?
- Suppose that the follower can appropriate the lower cost technology of the leader. Patent protection may affect the appropriation incentives by the follower.
 - Effect of strengthening protection on appropriation incentives for the follower.
 - Effect of strengthening protection on innovation incentives by the leader.
- How does the analysis change if the firms compete in prices?

References

- Mas-Colell- 12E